Google is committed to advancing racial equity for Black communities. See how.

Responsive layout grid

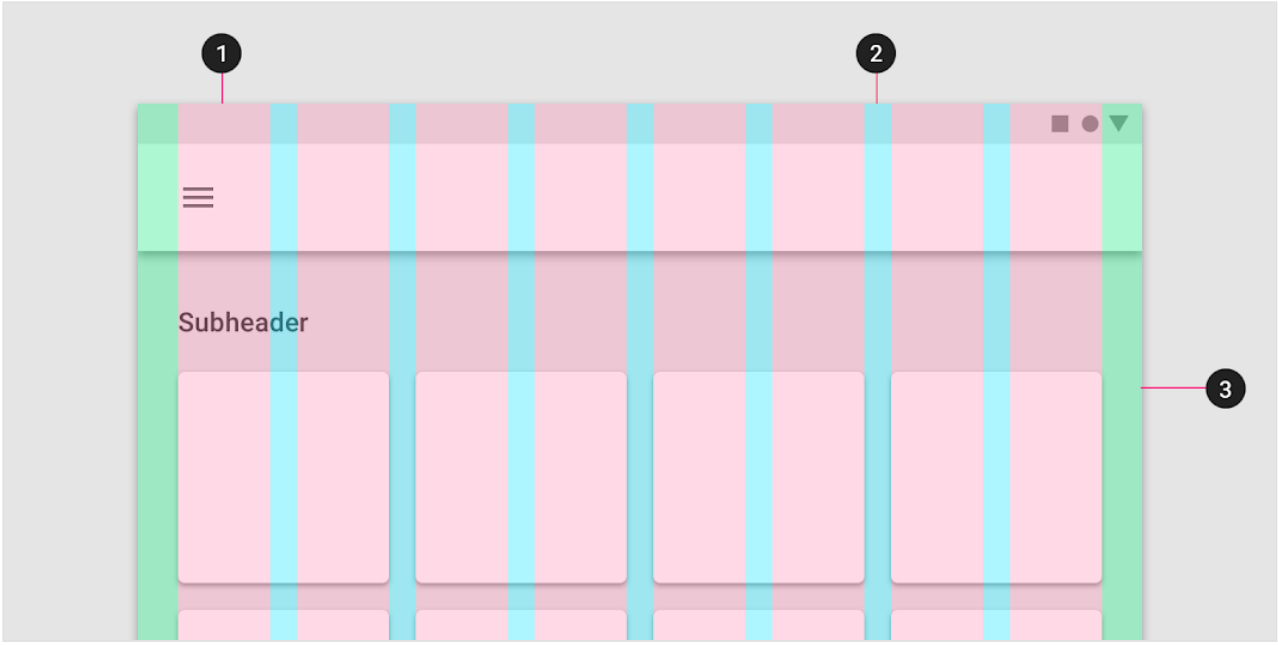
Material Design's responsive layout grid adapts to screen size and orientation, ensuring consistency across layouts.

CONTENTS

- **Columns, gutters, and margins**
- Grid customization
- Breakpoints
- Layout anatomy

Columns, gutters, and margins

The responsive layout grid is made up of three elements: columns, gutters, and margins.

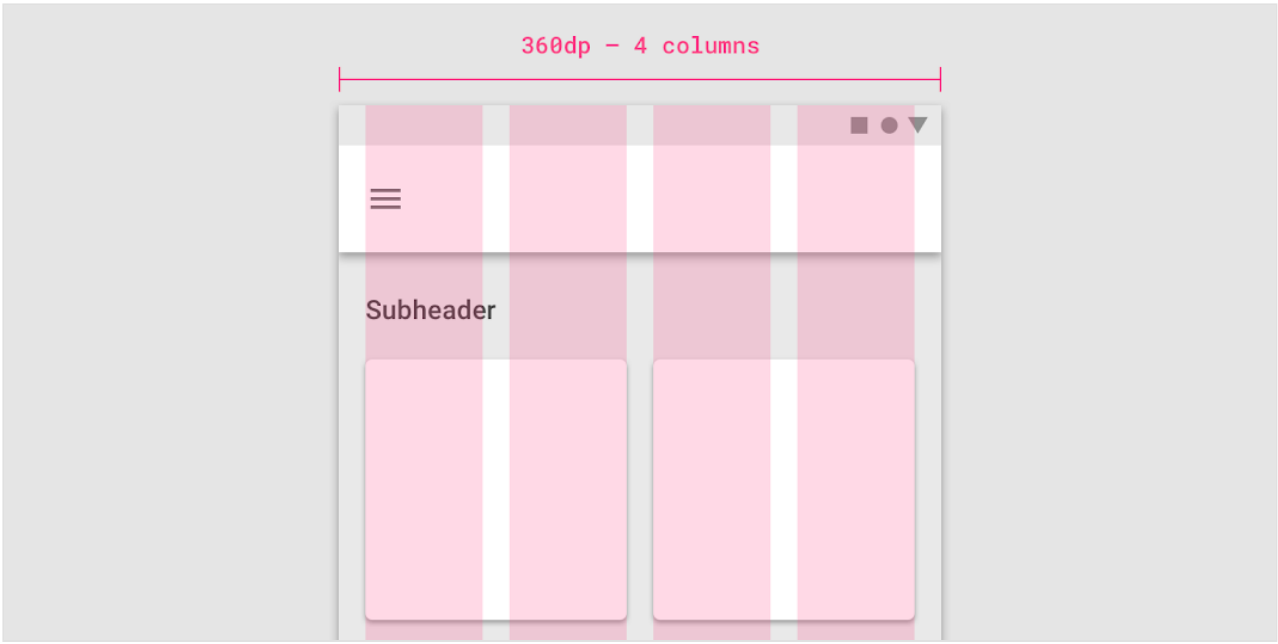


1. Columns
2. Gutters
3. Margins

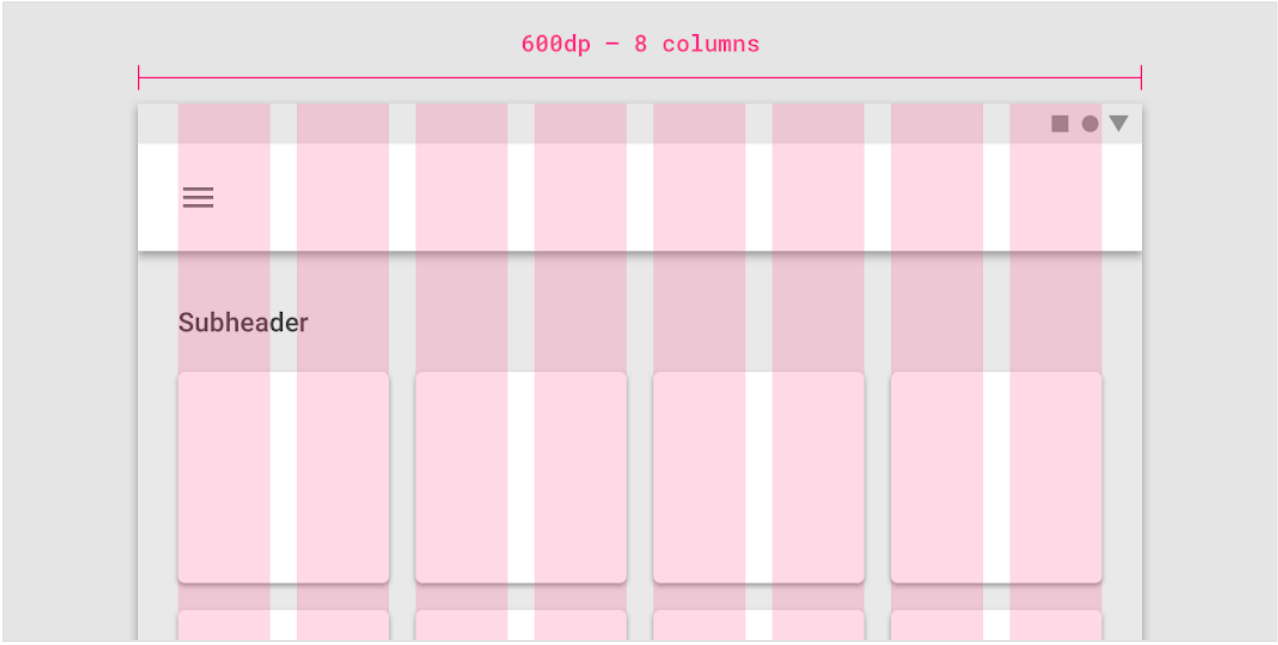
Columns

Content is placed in the areas of the screen that contain columns.

In responsive layouts, column width is defined with percentages, rather than fixed values. This allows content to adapt to any screen size. The number of columns displayed in the grid is determined by the breakpoint range, a range of predetermined screen sizes. A breakpoint can correspond with mobile, tablet, or other screen type.



On mobile, at a breakpoint of 360 dp, this layout grid uses 4 columns.



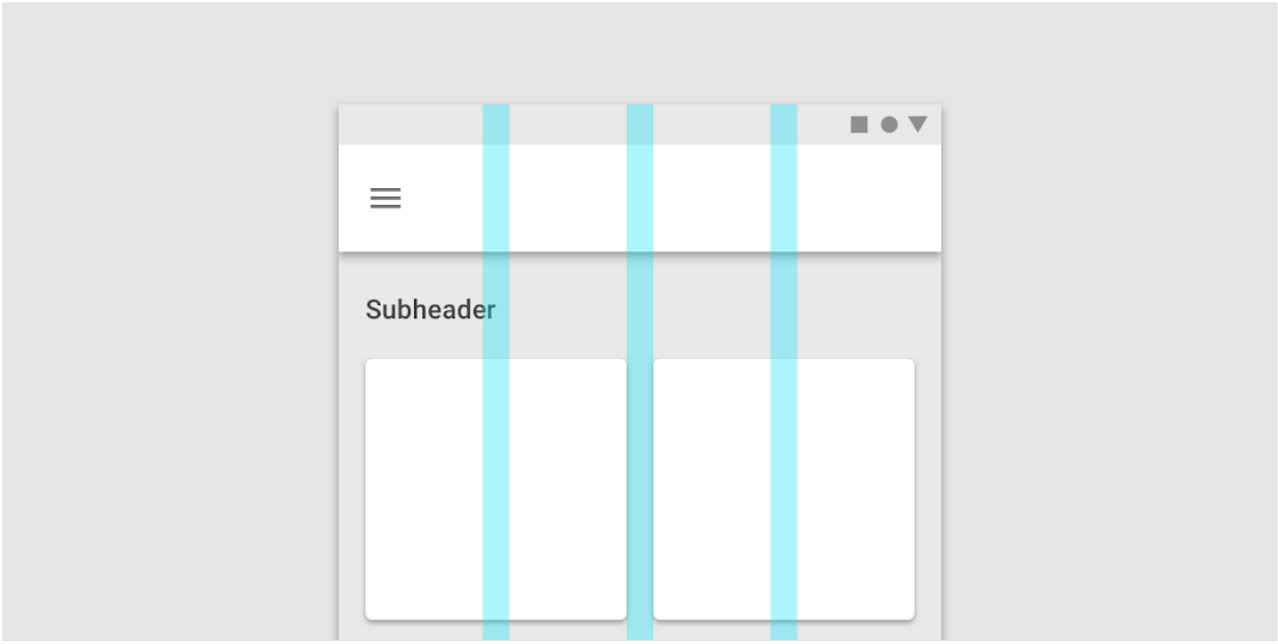
On tablet, at a breakpoint of 600 dp, this layout grid uses 8 columns.

Gutters

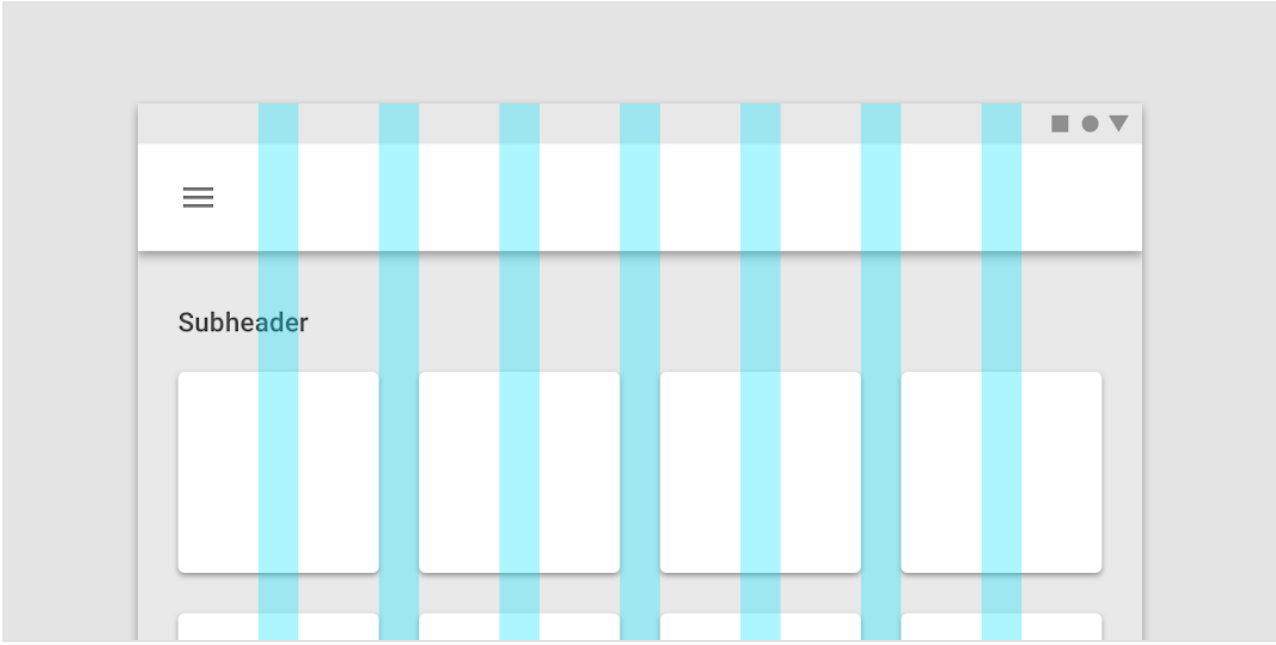
A gutter is the space between columns that helps separate content.

Gutter widths are fixed values at each breakpoint range. To better adapt to a given screen size, gutter widths can change at different breakpoints.

Wider gutters are more appropriate for larger screens, as they create more open space between columns.



On mobile, at a breakpoint of 360 dp, this layout grid uses 16dp gutters.

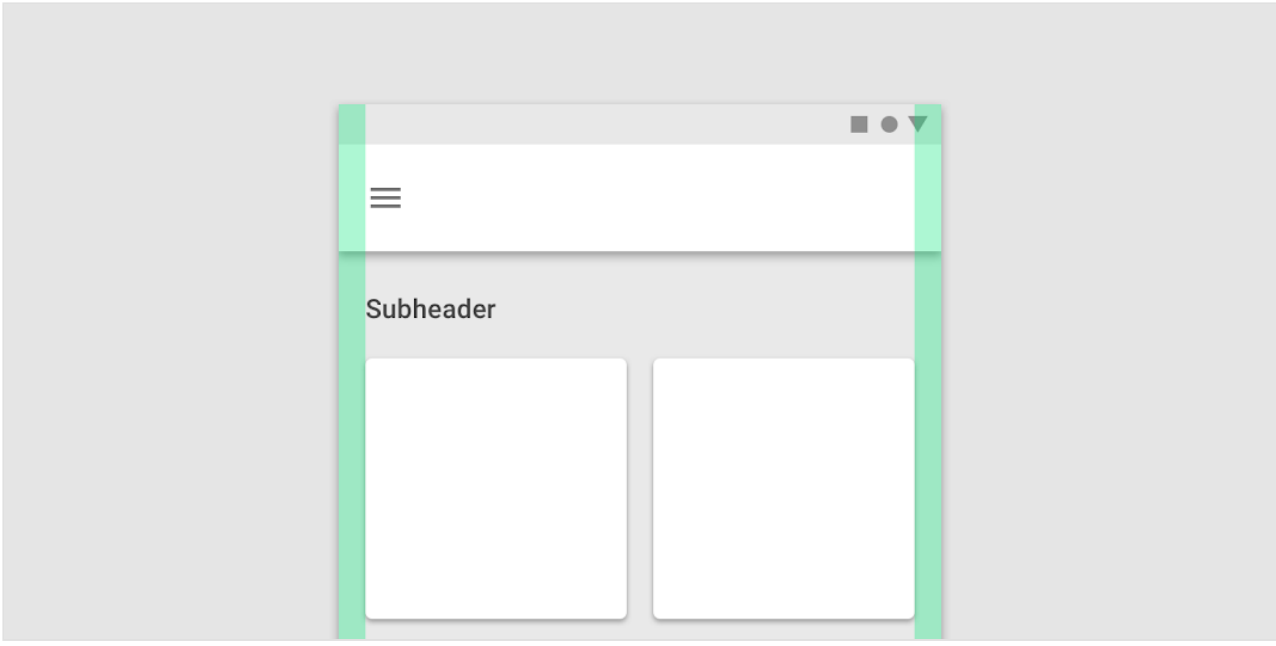


On a tablet, at a breakpoint of 600 dp, this layout grid uses 24dp gutters.

Margins

Margins are the space between content and the left and right edges of the screen.

Margin widths are defined using fixed or scaling values at each breakpoint range. To better adapt to the screen, the margin width can change at different breakpoints. Wider margins are more appropriate for larger screens, as they create more whitespace around the perimeter of content.



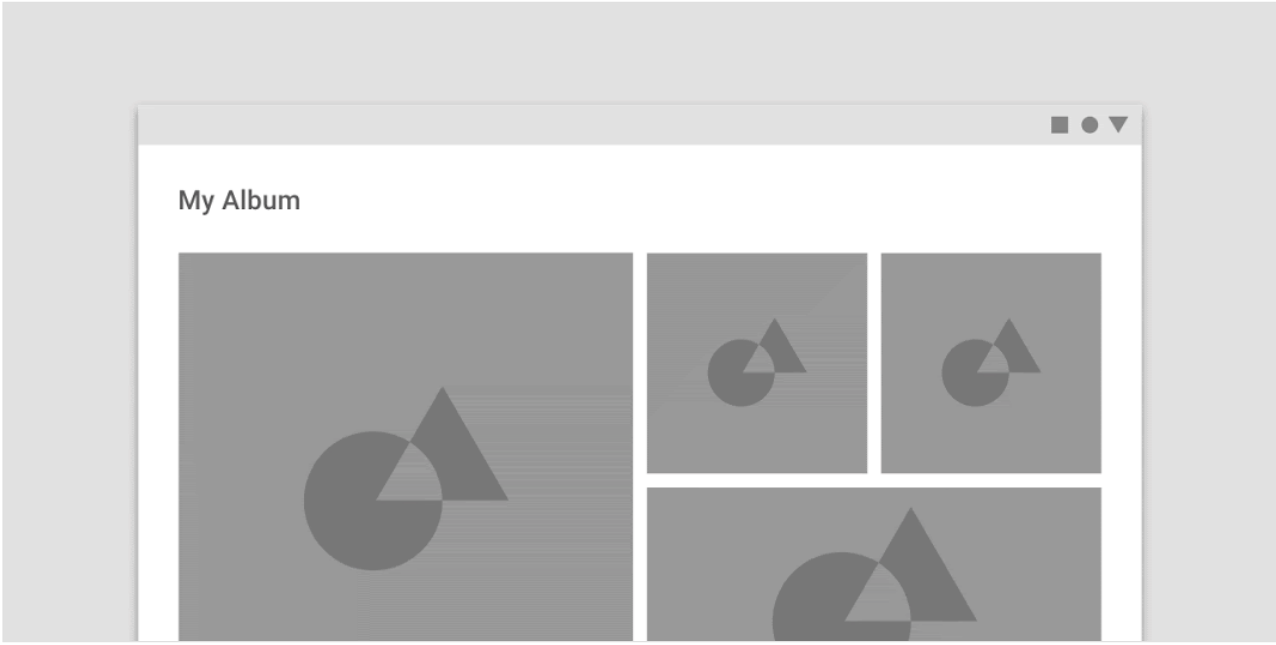
On mobile, at a breakpoint of 360 dp, this layout grid uses 16dp margins.

Grid customization

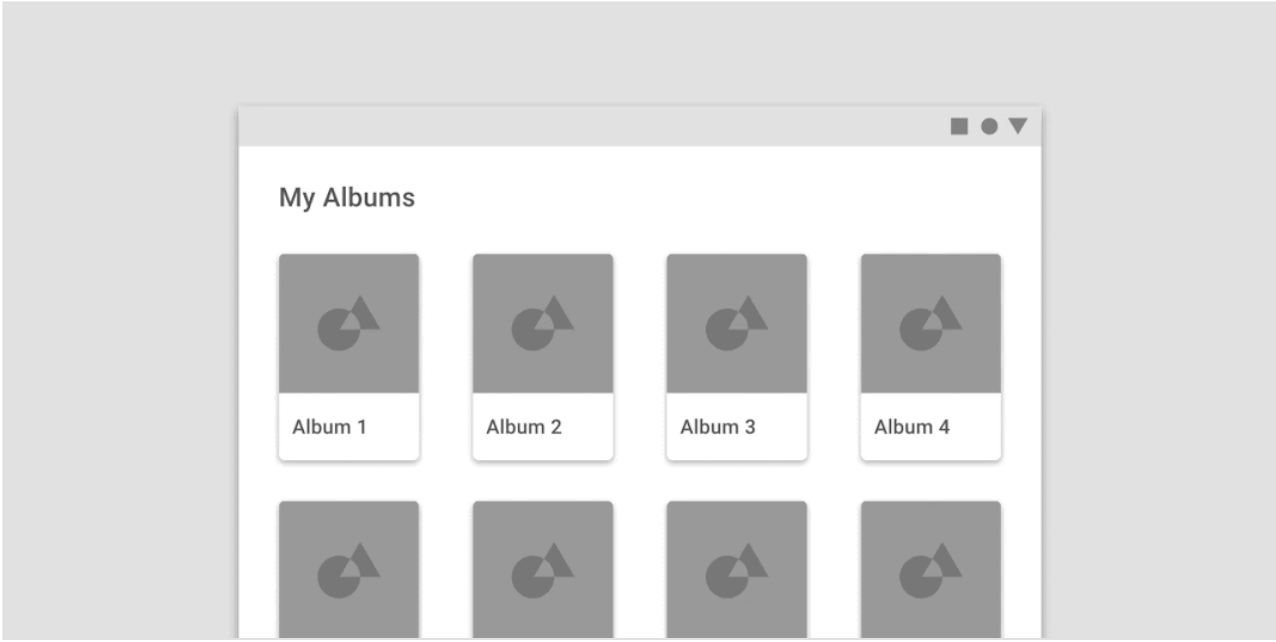
The layout grid can be adjusted to meet the needs of both your product and multiple device sizes.

Customizing gutters

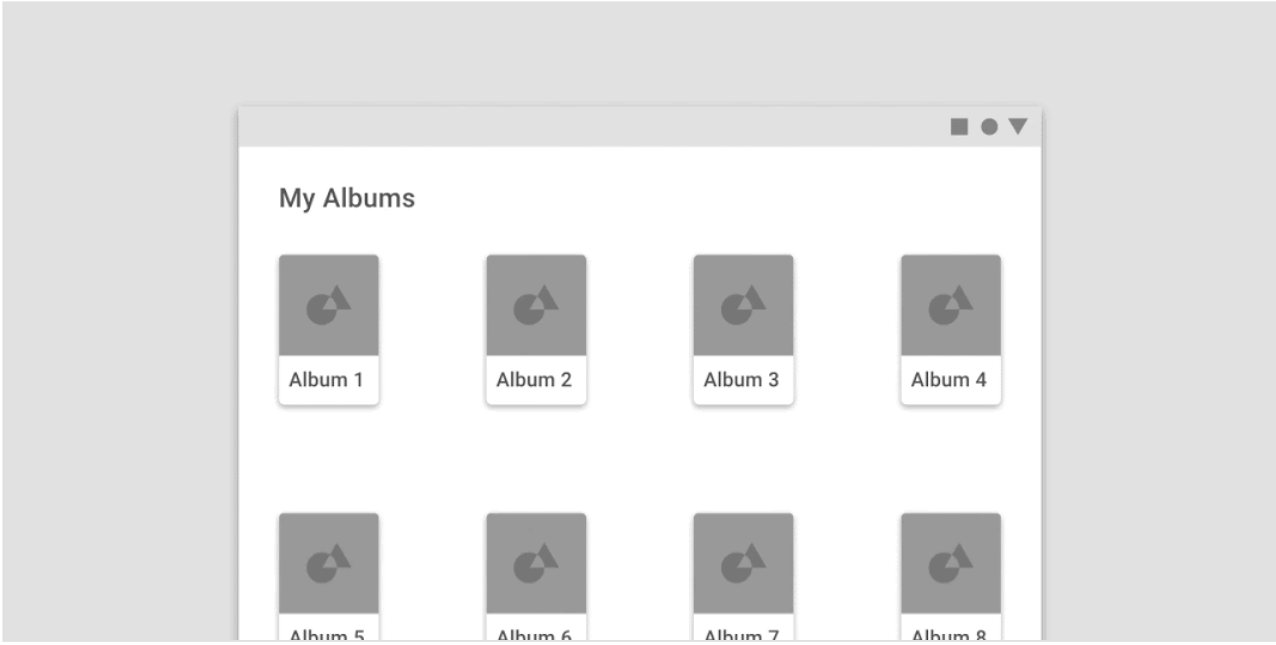
Gutters can be adjusted to create more or less space between the columns of a layout.



This layout grid uses 8dp gutters. The tighter spacing may suggest the images are closely related to one another, so that they are perceived as part of a collection.



This layout grid uses larger, 32dp gutters to create more separation between columns. The extra space helps each album to be perceived as an individual entity within a collection.

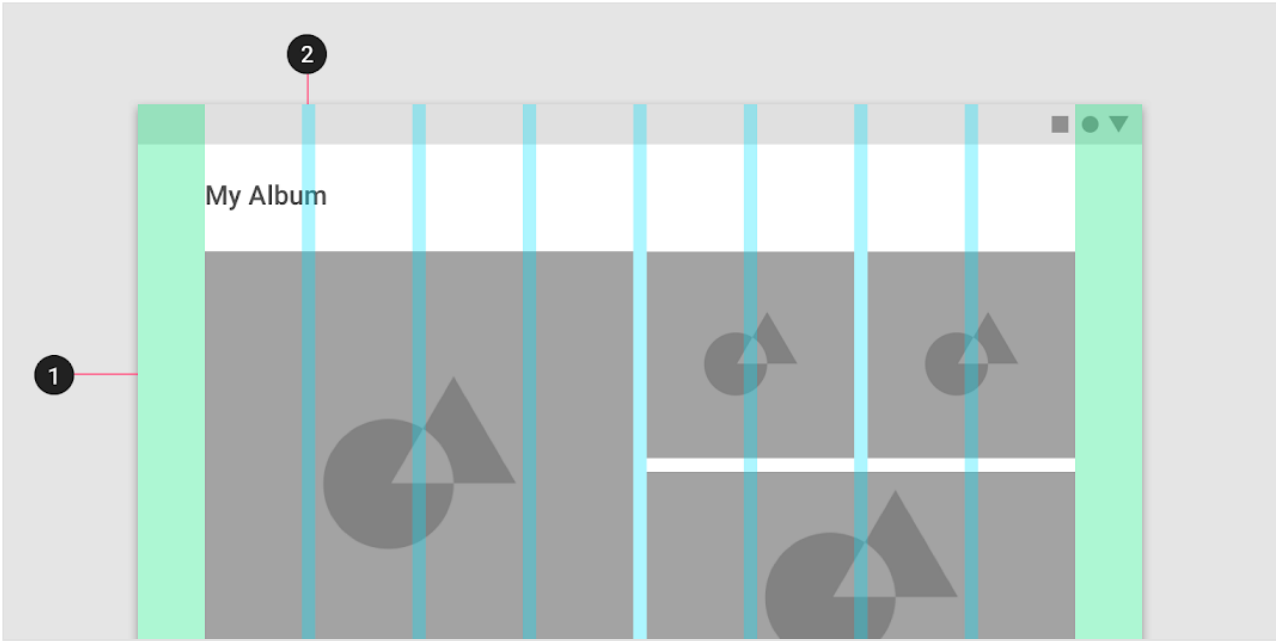


Don't

Don't make gutters too large or the same width as the columns. Oversized gutters won't leave enough room for content and may prevent a layout from appearing unified.

Gutters and margins

Within the same breakpoint, gutter and margin widths can be different from each other.

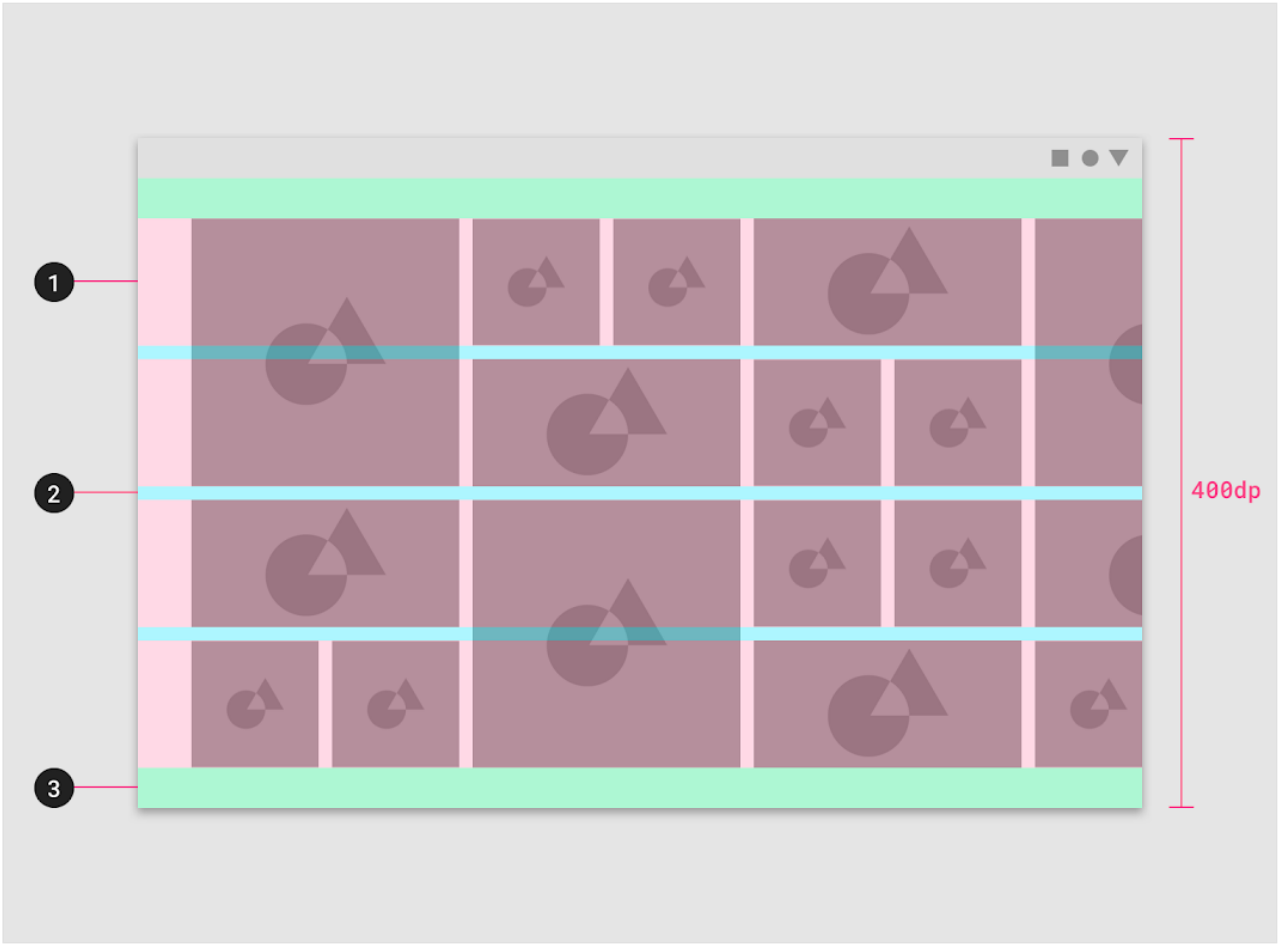


1. 32dp margins
2. 8dp gutters

Horizontal grids

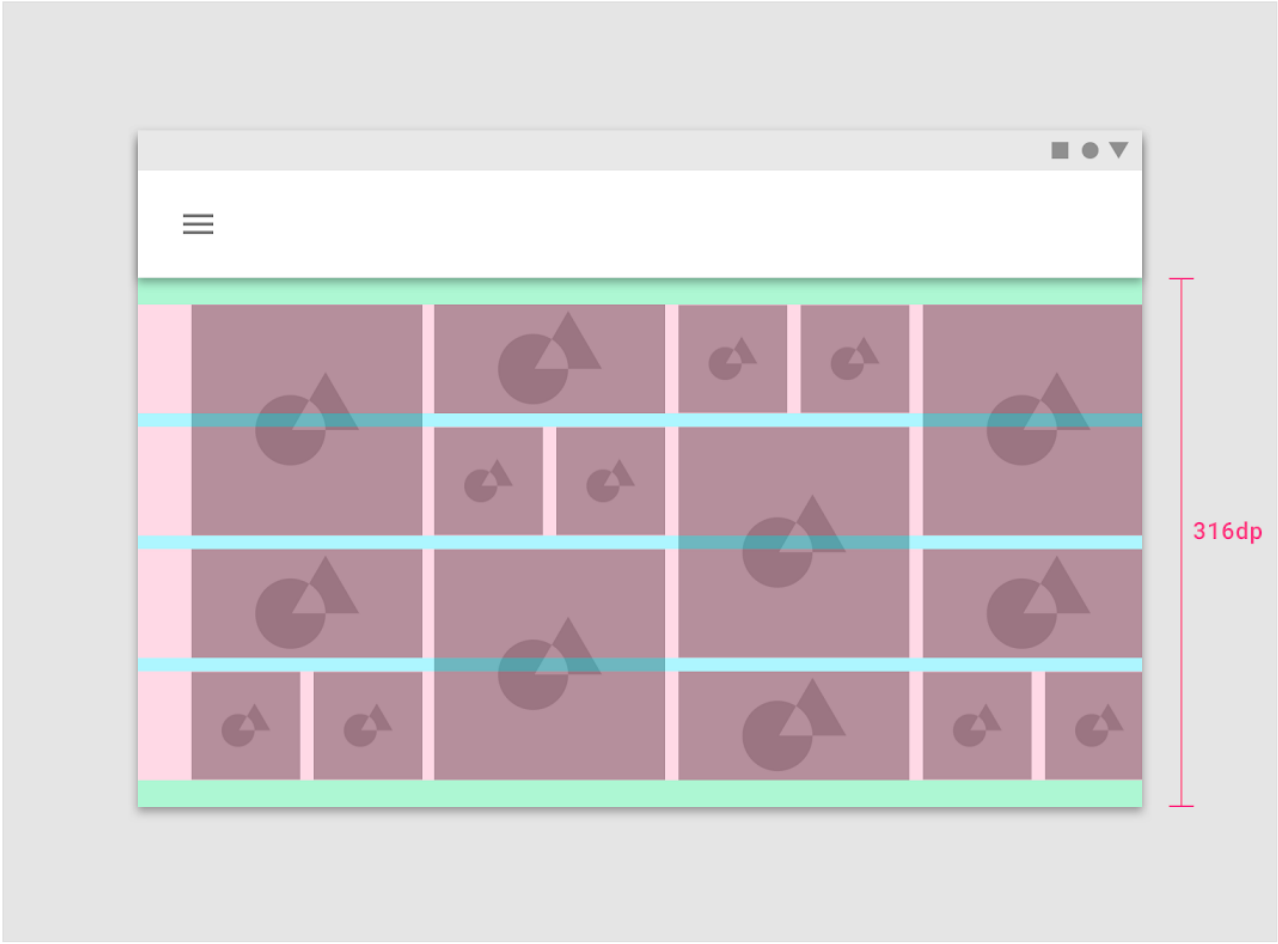
The Material Design layout grid can be customized for touch UIs that scroll horizontally. Columns, gutters, and margins are laid out from left to right, rather than top to bottom. The height of the screen determines the number of columns in a horizontal grid.

Horizontally scrolling UIs are uncommon on non-touch and web platforms.



This horizontal layout grid uses four horizontal columns, for a total layout height of 400dp.
1. Columns
2. Gutters
3. Margins

Horizontal grids can be positioned to accommodate different heights, allowing space for app bars or other UI regions at the top.



This horizontal layout grid starts below the Top App Bar component and uses four horizontal columns at a height of 316dp.

Top app bar

The top app bar displays…

Related article

Breakpoints

A breakpoint is the screen size threshold determined by specific layout requirements. At a given breakpoint range, the layout adjusts to suit the screen size and orientation.

Material Design provides responsive layouts based on 4-column, 8-column, and 12-column grids, available for use across different screens, devices, and orientations.

Each breakpoint range determines the number of columns, and recommended margins and gutters for each display size.

Breakpoint system

The responsive layout grid is primarily used to organize content and components in a layout's body region. Layout regions are described in detail in the *Understanding layout* guidance.

When scaling a layout for different screen sizes or orientations, the responsive grid adjusts margin and body widths as well as the number of columns in the layout.

Screen size	Margin	Body	Layout columns
Extra-small (phone)			
0-599dp	16dp	Scaling	4
Small (tablet)			
600-904	32dp	Scaling	8
905-1239	Scaling	840dp	12
Medium (laptop)			
1240-1439	200dp	Scaling	12
Large (desktop)			
1440+	Scaling	1040	12

At extra small breakpoints, margins have a baseline value of 16dp. The body remains responsive to increases in size. Upon reaching the next larger breakpoint, small at 600dp wide), the margin increases to 32dp.

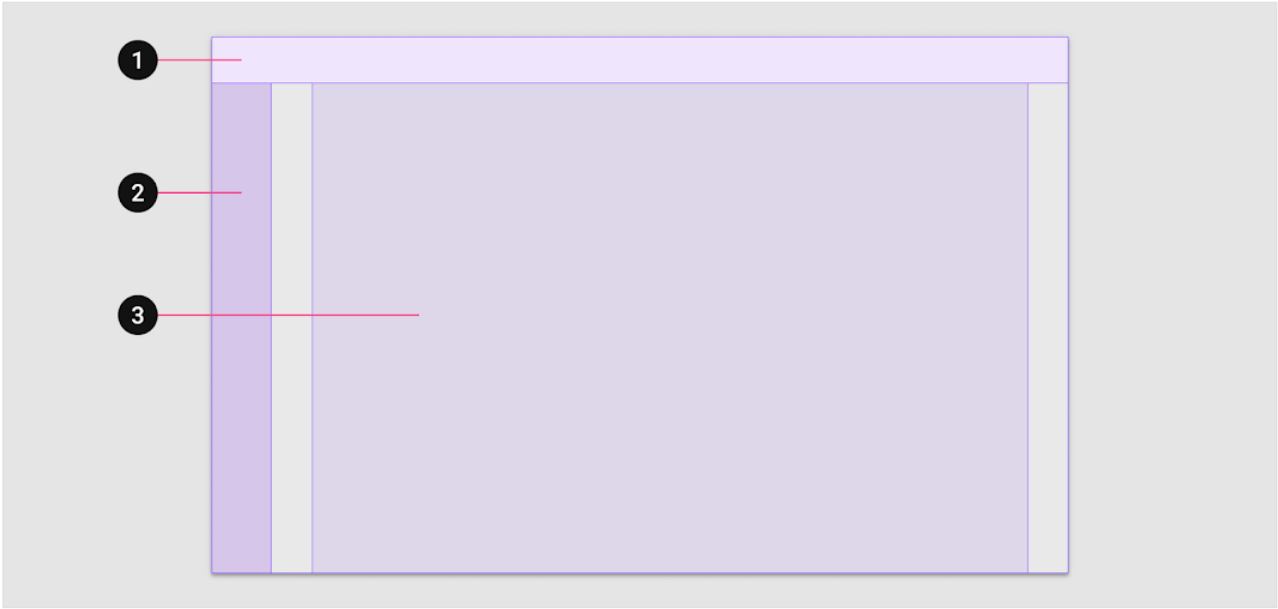
When the width of the body reaches 840dp, the margins become flexible and increase up to a maximum margin width of 200dp. After the maximum width is reached, the body region once again becomes responsive, continuing to scale horizontally. At large breakpoints, the body region can maintain a maximum width while margins again scale horizontally.

Layout anatomy

Layout regions are the foundation for a composition. They're building blocks for a layout and are composed of elements and components that share similar functions. Layout regions can also be composed of other smaller containers as well.

A large screen layout structure is based on three main elements:

- Body
- Navigation
- App bars



1. App bars 2. Navigation 3. Body

Material Design's responsive layout grid primarily informs the layout's body region. Each of the three layout regions are described in detail in the *Understanding layout* guidance.

Up next



Material is an adaptable system of guidelines, components, and tools that support the best practices of user interface design. Backed by open-source code, Material streamlines collaboration between designers and developers, and helps teams quickly build beautiful products.

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